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Tooling and method for assembling a vane ring

Abstract

Tooling and methods for assembling a vane ring of a turbine engine are provided. A tool includes a base configured to receive an outer ring and vanes radially inwardly of the outer ring. A heater is operable to apply heat to the outer ring when the outer ring is received on the base. Expandable jaws having a plurality of radially actuatable pads are operable to urge the vanes radially outwardly toward the outer ring. A press is operable to apply an axial force against an inner support of the vane ring to install the inner support radially inwardly of the vanes.

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Background/Summary

TECHNICAL FIELD

(1) The disclosure relates generally to assembling press-fitted components, and more particularly to tooling and method for assembling vane assemblies of turbine engines.

BACKGROUND

(2) Some structures of turbine engines are assembled using press fits. For press-fitted assemblies that have several components, existing assembly procedures can be expensive, time consuming and must be executed carefully to obtain the desired result and avoid damaging any of the components. Improvement is desired.

SUMMARY

(3) In one aspect, the disclosure describes a tool for assembling a vane ring including an outer ring, an inner support and a plurality of vanes disposed radially between the outer ring and the inner support. The tool comprises: a base configured to receive the outer ring and the vanes radially inwardly of the outer ring and angularly distributed about an axis; a heater operable to apply heat to the outer ring when the outer ring is received on the base; jaws having a plurality of radially actuatable pads operable to urge the vanes radially outwardly toward the outer ring; and a press operable to apply an axial force against the inner support to install the inner support radially inwardly of the vanes.

- (4) The base may include a receptacle configured to receive a boss on a vane segment including one or more of the vanes to angularly locate the vane segment on the base.
- (5) The receptacle may include a slot extending axially relative to the axis and having an open axial end for receiving the boss therein when the vane segment is received on the base by an axial movement of the vane segment relative to the base.
- (6) The base may include a locating feature configured to interact with the inner support to angularly locate the inner support on the base.
- (7) The heater may be configured apply heat to the outer ring by conductive heat transfer. The heater may extend along a majority of a circumference of the outer ring.
- (8) The press may include a drift defining an annular interface for applying the axial force against the inner support. The jaws may define a central opening that is sized to permit a passage of the drift therethrough.
- (9) The pads may be angularly distributed about the central opening of the jaws.
- (10) The jaws may include: a first ring configured to be secured to the base; and a second ring coaxial with the first ring and rotatable about the axis when the first ring is secured to the base. The second ring may be operatively connected to the pads to actuate the pads.
- (11) The tool may comprise a plurality of cams connected to the second ring of the jaws and rotatable with the second ring. The cams may be respectively engaged with follower surfaces of the pads to convert rotary motion of the cams into radially outward motion of the pads.
- (12) The tool may comprise a cam connected to the second ring of the jaws and rotatable with the second ring. The cam may be engaged with a follower surface of a first of the pads to convert rotary motion of the cam into radial motion of the first pad.
- (13) The cam may be located radially inwardly of the first pad.
- (14) The first pad may be engaged with the first ring via a guide extending radially relative to the axis.
- (15) The guide may include a rail secured to the first ring. The guide may be slidably received in a slot defined in the first pad.
- (16) Embodiments may include combinations of the above features.
- (17) In another aspect, the disclosure describes jaws for urging vanes radially outwardly toward an outer ring during assembly of a vane ring. The jaws comprise: a stationary ring configured to be secured to a base, the stationary ring having a central axis extending therethrough; a movable ring movably mounted to the stationary ring and being rotatable relative to the stationary ring about the central axis; and a plurality of radially movable pads, the pads being operatively connected to the movable ring and actuatable by movement of the movable ring relative to the stationary ring.
- (18) The jaws may comprise a plurality of cams connected to the movable ring and rotatable with the movable ring about the axis. Each of the cams may be engaged with a follower surface of one of the pads to convert rotary motion of the movable ring into radial motion of the pads.
- (19) The stationary ring may be disposed axially between the movable ring and the pads. The cams may be connected to the movable ring via a central opening of the stationary ring.
- (20) The jaws may comprise a plurality of rails secured to the stationary ring and extending radially. The rails may be angularly spaced apart around the stationary ring. The pads may each include a slot formed therein for engagement with a respective one of the rails.
- (21) Embodiments may include combinations of the above features.
- (22) In a further aspect, the disclosure describes a method of assembling a vane ring of a turbine engine. The method comprises: receiving an outer ring and a plurality of vane segments of the vane ring on a tooling base, the vane segments being disposed radially inwardly of the outer ring; heating the outer ring; while the outer ring is being heated: urging the vane segments radially outwardly toward the outer ring; partially receiving an inner support of the vane ring radially inwardly of the vane segments; and after urging the vane segments radially outwardly toward the outer ring, applying an axial force against the inner support to install the inner support radially

inwardly of the vane segments and establish a press fit between the outer ring, the vane segments and the inner support.

(23) The inner support may have a seal installed thereon and extending around the inner support. Applying an axial force against the inner support may include radially compressing the seal between the inner support and the vane segments.

(24) The seal may be a first seal. The method may include: receiving a second seal between the vane segments and the outer ring; and radially compressing the second seal between the vane segments and the outer ring.

(25) Embodiments may include combinations of the above features.

(26) Further details of these and other aspects of the subject matter of this application will be apparent from the detailed description included below and the drawings.

Description

DESCRIPTION OF THE DRAWINGS

(1) Reference is now made to the accompanying drawings, in which:

(2) FIG. 1 shows an axial cross-section view of an exemplary turbine engine including a vane ring;

(3) FIG. 2 is a perspective view of a tool for assembling the vane ring of the engine of FIG. 1;

(4) FIG. 3 is a perspective view of a base of the tool of FIG. 2;

(5) FIG. 4 is a perspective cross-section view of part of the base of the tool taken along line 4-4 in FIG. 3 showing part of the vane ring being received on the base;

(6) FIG. 5 is a perspective cross-section view of part of the base of the tool taken along line 4-4 in FIG. 3 showing another part of the vane ring being received on the base;

(7) FIG. 6A is a perspective view showing a heater being installed onto an outer ring of the vane ring after the outer ring has been received on the base of the tool;

(8) FIG. 6B is a perspective view of the heater installed onto the outer ring of the vane ring before receiving the outer ring on the base of the tool;

(9) FIG. 7A is a perspective view showing jaws being installed onto the base of the tool;

(10) FIG. 7B is a perspective view showing a connection between a manual lever and the jaws for actuating the jaws;

(11) FIG. 8 is a perspective view of the jaws showing movable pads of the jaws;

(12) FIG. 9 is another perspective view of the jaws showing a movable ring of the jaws;

(13) FIGS. 10A and 10B are schematic top plan views of one of the pads of the jaws actuatable using a cam and shown in a contracted configuration of the jaws and in an expanded configuration of the jaws respectively;

(14) FIG. 11 is a schematic top plan view of one of the pads of the jaws actuatable using a linkage;

(15) FIG. 12 is a perspective view of a drift being passed through the jaws to access an inner support of the vane ring;

(16) FIG. 13 is a perspective cross-section view of the tool of FIG. 2 and components of the vane ring taken along line 13-13 in FIG. 2; and

(17) FIG. 14 is a flow diagram of a method of assembling a vane ring of a turbine engine.

DETAILED DESCRIPTION

(18) This disclosure describes tooling and associated methods for facilitating the assembly of vane rings for turbine engines. In some embodiments, the tools and methods described herein may facilitate the assembly of vane rings that include a plurality of components that are press-fitted together. In some embodiments, the tools and methods described herein may promote ease of use and reduce the risk of damaging the components of the vane ring during assembly. For example, the tools and methods described herein may promote desirable relative positioning of the components of the vane ring when the components are press-fitted together. Aspects of various

embodiments are described through reference to the drawings.

(19) The terms “connected” and “secured” may include both direct connection and securement (in which two components contact each other) and indirect connection and securement (in which at least one additional component is located between the two components). The term “coaxial” as used herein is intended to encompass geometric deviations (e.g., within intended dimensional tolerances) that do not result in a change in the basic function to which it is related. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.

(20) FIG. 1 illustrates an exemplary (e.g., gas) turbine engine **10** (referred hereinafter as “engine **10**”) of a type preferably provided for propelling an aircraft in subsonic flight. Engine **10** as illustrated in FIG. 1 is a turbofan engine but engine **10** may alternatively be a turboprop engine or a turboshaft engine. In some embodiments, engine **10** may be an auxiliary power unit that is used onboard an aircraft for functions other than propulsion. In some embodiments, engine **10** may be of a type suitable for ground-based (e.g., industrial) applications.

(21) Engine **10** may include, in serial flow communication, fan **12** through which ambient air is propelled, multistage compressor **14** for pressurizing the air, combustor **16** in which the compressed air is mixed with fuel and ignited for generating an annular stream of hot combustion gases, and turbine section **18** for extracting energy from the combustion gases. Compressor **14**, combustor **16** and turbine section **18** may be in serial fluid communication via core gas path **20** of engine **10**. Core gas path **20** may be disposed radially inward of bypass duct **22** of engine **10**.

(22) Engine **10** may include one or more vane rings **24** (referred hereinafter in the singular) for directing a flow of gas (e.g., air, combustion gas) through core gas path **20** of engine **10**. Vane ring **24** may include a plurality of fixed-orientation (i.e., non-variable) stator guide vanes that are angularly distributed about central axis CA and that are stationary within core gas path **20**. Central axis CA may correspond to an axis of rotation of a rotor of compressor **14** and/or an axis of rotation of a rotor of turbine section **18**. In some embodiments of engine **10**, vane ring **24** may be installed upstream of combustor **16** and within compressor **14** to guide compressed air for example. In some embodiments of engine **10**, vane ring **24** may be installed downstream of combustor **16** and within turbine section **18** to guide combustion gas for example. Vane ring **24** may be made from one or more suitable (e.g., metallic) material such as (e.g., stainless) steel or a nickel-based alloy for example.

(23) FIG. 2 is a perspective view of an exemplary tool **26** for assembling vane ring **24** of engine **10** or another vane ring. In some embodiments, tool **26** may facilitate the assembly of vane ring **24**, which may include a plurality of components that are press-fitted together. In the configuration of tool **26** shown in FIG. 2, the components of vane ring **24** may be disposed inside of tool **26** and may not be readily visible from the vantage point of FIG. 2. In some embodiments, tool **26** may promote desirable relative positioning of components and ease of use to reduce the risk of damaging components of vane ring **24** during assembly of vane ring **24**. Tool **26** may include base **28** configured to receive components of vane ring **24** (e.g., individually or in groupings) in preparation for assembly of vane ring **24**. Tool **26** may include (e.g., electric) heater **30** operable to apply heat to and cause thermal expansion of one or more components of vane ring **24** during the assembly of vane ring **24**.

(24) Tool **26** may include actuatable jaws **32** configured to urge one or more components of vane ring **24** radially outwardly (i.e., apply a pre-load on some components) during the assembly of vane ring **24**. Jaws **32** may be actuatable from a contracted configuration C to an expanded configuration E by the actuation of manual lever **34**. Alternatively, jaws **32** may be actuated by way of an electric, pneumatic or hydraulic actuator for example. The use of jaws **32** may promote desirable relative positioning of components of vane ring **24** during press fitting.

(25) Tool **26** may include press **36** operable to apply an axial force F1 downwardly along central axis CA against one or more components of vane ring **24** to press fit the one or more components

together during the assembly of vane ring 24. Press 36 may include drift 38 defining an interface for applying axial force F1 against the one or more components of vane ring 24. Jaws 32 may define a central opening 40 that is sized to permit a passage of drift 38 therethrough. Press 36 may include (e.g., hydraulic) actuator 42 such as a hydraulic cylinder to generate axial force F1 that is applied to one or more components of vane ring 24 via drift 38. In various embodiments, actuator 42 of press 36 may include an electric or pneumatic actuator to generate axial force F1 for example. Other components of tool 26 and the operation of tool 26 for the assembly of vane ring 24 are described below.

(26) FIG. 3 is a perspective view of base 28 of tool 26. Base 28 may receive and hold components of vane ring 24 and components of tool 26 to facilitate the assembly of vane ring 24. Base 28 may include platform 44 onto which other components of base 28 are secured (e.g., fastened). Base 28 may include hub 46 secured to base 28 and configured to receive outer ring 48 of vane ring 24 and optionally also vane segments 50 (shown in FIG. 4) of vane ring 24. Hub 46 may define an interface for receiving outer ring 48 and vane segments 50 thereon in order to locate outer ring 48 on hub 46. In some embodiments, hub 46 may have a geometry that at least partially conforms to the geometry of outer ring 48 so that outer ring 48 may be seated and retained on hub 46 during assembly of vane ring 24. For example, a cylindrical surface of hub 46 may be received inside of outer ring 48 and engage (e.g., mate) with a cylindrical surface of outer ring 48. In some embodiments, hub 46 may have a generally annular shape.

(27) Base 28 may include guide plate 52, which may include a generally cylindrical projection 53 extending axially upwardly and away relative to platform 44. Cylindrical projection 53 of guide plate 52 may extend through a central opening of hub 46. In other words, cylindrical projection 53 of guide plate 52 may be disposed radially inwardly of hub 46. Guide plate 52 may be secured to platform 44 or may be movably mounted to platform 44. For example, guide plate 52 may be axially movable along central axis CA. In some embodiments, movement of guide plate 52 transverse to central axis CA may be prevented, and angular movement of guide plate 52 about central axis CA may also be prevented. As explained further below, guide plate 52 may assist with the positioning of vane segments 50 between guide plate 52 and outer ring 48. Guide plate 52 may also assist with the installation of inner support 54 (shown in FIG. 5) of vane ring 24.

(28) Base 28 may include central rod 56 extending coaxially with central axis CA. Central rod 56 may be secured (e.g., fastened) to platform 44. Central rod 56 may serve to guide the axial movement of drift 38 along central axis CA during the application of axial force F1 onto inner support 54 for example. In some embodiments, central rod 56 and hub 46 may be coaxial. In some embodiments, central rod 56, hub 46 and cylindrical projection 53 of guide plate 52 may be coaxial.

(29) Base 28 may include one or more (e.g., three) posts 58A-58C secured (e.g., fastened) to platform 44 and extending upwardly from platform 44. Post(s) 58A-58C may be used to secure part of jaws 32 to platform 44. Posts 58A-58C may be angularly spaced apart about central axis CA and may be secured to platform 44 at respectively locations that are radially outward of hub 46.

(30) Manual lever 34 may be secured (e.g., fastened) to platform 44 via spacer block 60 so that manual lever 34 may be raised from platform 44. Manual lever 34 may be operatively connected to push/pull rod 62 so that the actuation of manual lever 34 may correspondingly cause a linear actuation of push/pull rod 62 along arrow 63. For example, a movement of manual lever 34 toward expanded configuration E may cause an extension of push/pull rod 62, and a movement of manual lever 34 toward contracted configuration C may cause a retraction of push/pull rod 62.

(31) FIG. 4 is a perspective cross-sectional view of part of base 28 of tool 26 taken along line 4-4 in FIG. 3. FIG. 4 also shows part of vane ring 24 that is sectioned along the same line 4-4. Vane ring 24 may include outer ring 48 and a plurality of vane segments 50 that are disposed radially inwardly of outer ring 48 and that are angularly distributed about central axis CA. Each vane segment 50 may include one or more stator vanes 64 that interact with the flow of gas in core gas

path **20** of engine **10** and that change the orientation of the flow of gas in core gas path **20**. As illustrated in FIG. **4** adjacent vane segments **50** may interface with (e.g., contact) one another along an interface that is oriented obliquely relative to central axis CA. The number of vane segments **50** and the number of stator vanes **64** per vane segment **50** may vary depending on the size of and configuration of vane ring **24**. In the example shown in FIG. **4**, vane ring **24** may include seven separate vane segments **50**.

(32) Depending on the configuration of outer ring **48** and vane segments **50**, outer ring **48** and vane segments **50** (and optionally also seal **118** shown in FIG. **13**) may be pre-assembled together outside of tool **26** and then received onto base **28** together as a unit. As shown in FIG. **4**, outer ring **48** and vane segments **50** may be lowered as a unit onto base **28** along central axis CA.

Alternatively, outer ring **48**, vane segments **50** and optionally also seal **118** may be individually installed on base **28**. As outer ring **48** is lowered onto hub **46**, one or more surfaces of outer ring **48** may engage with hub **46** so that hub **46** may cause outer ring **48** to become coaxial with hub **46** and with central axis CA. In some embodiments, a radially outer surface of hub **46** may be axially tapered so that when the radially outer surface of hub **46** is received inside part of outer ring **48**, the gradual lowering of outer ring **48** onto hub **46** will cause outer ring **48** to gradually become aligned with hub **46** and also become seated onto hub **46**.

(33) Guide plate **52** may assist with locating vane segments **50** around central axis CA. For example, guide plate **52** and vane segments **50** may include cooperating locating features that allow angular alignment (i.e., locating) of vane segments **50** around central axis CA. In other words, the locating features on guide plate **52** and vane segments **50** may allow vane segments **50** to be clocked about central axis CA. In some embodiments, such locating features may include one or more receptacles **66** defined in guide plate **52** and configured to receive respective bosses **68** (i.e., protrusions) defined on one or more vane segments **50**. In some embodiments, each vane segment **50** may include one or more bosses **68**, and guide plate **52** may include a corresponding number of receptacles **66**.

(34) In some embodiments, receptacle(s) **66** may be defined in cylindrical projection **53** of guide plate **52**, and boss(es) **68** may project radially inwardly from one or more radially-inward-facing surfaces of vane segment(s) **50**. In some embodiments, receptacle(s) **66** may include one or more slots extending axially relative to central axis CA and each having an open axial end for receiving a respective boss **68** therein when vane segment(s) **50** is/are lowered along central axis CA and received on base **28** by an axial movement of vane segment(s) **50** relative to base **28**.

(35) In some embodiments, one or more receptacles may instead be defined in one or more vane segments **50** and one or more corresponding bosses may be defined on guide plate **52**. In some embodiments, guide plate **52** may include one or more receptacles and one or more bosses, and vane segments **50** may include corresponding one or more receptacles and one or more bosses. It is understood that other types of angular locating features (e.g., pin, hole, key and key way) between vane segments **50** and guide plate **52** may be suitable.

(36) FIG. **5** is a perspective cross-sectional view of part of base **28** of tool **26** taken along line 4-4 in FIG. **3** showing outer ring **48** and vane segments **50** being received onto hub **46** where vane segments **50** are disposed radially between cylindrical projection **53** and outer ring **48**. FIG. **5** also shows inner support **54** and seal **70** that are sectioned along the same line 4-4. Inner support **54** may have a generally annular shape and may define a central hub of vane ring **24**. Vane ring **24** may include seal **70** that is compressed (i.e., press-fitted) between inner support **54** and vane segments **50**. Seal **70** may have a generally annular shape extending around central axis CA and may be pre-installed onto the exterior of inner support **54** so that inner support **54** and seal **70** may be received onto base **28** as a unit. Seal **70** may be made from a metallic material and may have a corrugated (e.g., W-shaped) cross-sectional profile. FIG. **5** shows the process of receiving inner support **54** and seal **70** onto base **28** by lowering inner support **54** and seal **70** together along central axis CA. Installing inner support **54** onto base **28** may include partially receiving inner support **54**

radially inwardly of vane segments **50**.

(37) Guide plate **52** and inner support **54** may include cooperating locating features that allow (e.g., angular) alignment (i.e., locating) of inner support **54** around central axis CA. In some embodiments, the locating features on guide plate **52** and inner support **54** may allow inner support **54** to be clocked about central axis CA and also be clocked relative to the previously clocked vane segments **50**. In some embodiments, such locating features may include one or more pins **72** integrally formed with guide plate **52** or otherwise secured (e.g., fastened) to guide plate **52**, and configured to be received in one or more corresponding holes **74** defined in inner support **54**. It is understood that other types of locating features (e.g., bosses, slots) between inner support **54** and guide plate **52** may be suitable.

(38) In some embodiments, guide plate **52** may be movable along one or more axial guides **76** (e.g., pins) and be spring-loaded via one or more springs **78** disposed between guide plate **52** and platform **44**. Accordingly, guide plate **52** may be movable axially along central axis CA to accommodate axial movement of inner support **54** during installation of inner support **54** while maintaining the desired angular orientation of inner support **54**. For example, inner support **54** may be manually lowered toward base **28** to partially be received radially inwardly of vane segments **50**. However, when axial force **F1** is applied to inner support **54** via drift **38**, inner support **54** may undergo additional axial displacement to complete the installation of inner support **54**. Accordingly, guides **76** and springs **78** may allow guide plate **52** to maintain engagement with inner support **54** and move downwardly with inner support **54** while inner support **54** is being installed.

(39) FIG. 6A is a perspective view showing heater **30** being installed onto outer ring **48** of vane ring **24** after outer ring **48**, vane segments **50** and inner support **54** have been received on base **28** of tool **26**. Heater **30** may be configured to apply heat to outer ring **48** by conductive heat transfer. Heater **30** may extend along a majority or substantially an entirety of the outer circumference of outer ring **48**. Heater **30** may be installed by lowering heater **30** toward base **28** and along central axis CA. Heater **30** may be installed on an exterior of outer ring **48** and placed in contact with a radially outer surface of outer ring **48**. Heater **30** may have a generally annular shape that may be expanded to allow heater **30** to be installed around outer ring **48** and contracted to establish physical contact with outer ring **48** to permit conductive heat transfer therewith. The expansion and contraction of heater **30** may be achieved by the opening and closing of clasp **80**.

(40) Heater **30** may be electrically powered and may include a resistive heating element. Heater **30** may be electrically connected to an electric power source via cable **82** and connector **84**. Heater **30** may include one or more handles **86** to facilitate the installation and removal of heater **30**.

(41) FIG. 6B is a perspective view of heater **30** installed onto outer ring **48** of vane ring **24** before installing outer ring **48** on base **28** of tool **26**. For example, heater **30**, outer ring **48** and optionally vane segments **50** and seal **118** (shown in FIG. 13) may be pre-assembled together outside of tool **26** and then received onto base **28** together as a unit.

(42) FIG. 7A is a perspective view showing jaws **32** being installed onto base **28** of tool **26** after outer ring **48**, vane segments **50**, inner support **54** and heater **30** have been received on base **28**. Jaws **32** may be installed by lowering jaws **32** toward base **28** and along central axis CA. Jaws **32** may serve as a lid containing the components of vane ring **24** inside tool **26**. Jaws **32** may define central opening **40** extending therethrough. Jaws **32** may have a generally annular shape and may be installed to be coaxial with central axis CA. In other words, central axis CA may extend through central opening **40** defined through jaws **32**. Jaws **32** may include stationary ring **88** configured to be secured to base **28**. Stationary ring **88** may include one or more perforated tabs **90A-90C** extending radially outwardly. Stationary ring **88** may be secured to base **28** by way of fastening tabs **90A-90C** to respective posts **58A-58C** using respective threaded fasteners **92A-92C** (shown in FIG. 2).

(43) Jaws **32** may include movable ring **94** movably mounted to stationary ring **88** and being rotatable about central axis CA relative to stationary ring **88**. Movable ring **94** may be rotatable

about central axis CA within a prescribed angular range. In some embodiments, movable ring **94** may be fastened to stationary ring **88** by way of one or more threaded fasteners **95** and slotted holes defined in movable ring **94** so that the length of the slotted holes may define the angular range of motion of movable ring **94** relative to stationary ring **88**. Jaws **32** may include one or more radially movable pads **96**. Pads **96** may be operatively connected to movable ring **94** and actuatable by movement of movable ring **94** relative to stationary ring **88**. When installed on base **28**, pads **96** of jaws **32** may be used to urge vane segments **50** radially outwardly toward outer ring **48** during assembly of vane ring **24** as explained below.

(44) FIG. 7B is an enlarged perspective view showing connector **98** between manual lever **34** and jaws **32** to permit jaws **32** to be actuated using manual lever **34**. Connector **98** may drivingly connect push/pull rod **62** to movable ring **94**. For example, movement of push/pull rod **62** along arrow **63** may be orientated to be at least partially tangential to central axis CA, which may correspond to the axis of rotation of movable ring **94**. Accordingly, the actuation of manual lever **34** may be converted to rotational motion of movable ring **94** along arrow **108**. Movement of manual lever **34** toward contracted configuration C may cause rotation of movable ring **94** in one direction and movement of manual lever **34** toward expanded configuration E may cause rotation of movable ring **94** in the opposite direction.

(45) FIG. 8 is a perspective view of jaws **32** in isolation. FIG. 8 shows a lower side of jaws **32** on which movable pads **96** are movably mounted. Pads **96** may be angularly distributed about central opening **40** of jaws **32** and about central axis CA. The number and positioning of pads **96** may correspond to the number and positioning of vane segments **50** so that each pad **96** may be associated with a corresponding vane segment **50**. The securing of jaws **32** to posts **58A-58C** may set the angular positioning of jaws **32** and may result in the angular positioning of pads **96** to be in register with the angular positioning of vane segments **50** previously set via receptacle(s) **66** and boss(es) **68**. Accordingly, each pad **96** may be actuatable to urge a corresponding vane segment **50** radially outwardly toward outer ring **48**.

(46) Stationary ring **88** may be disposed axially between movable ring **94** and pads **96**. Pads **96** may be movably engaged with stationary ring **88** so that rotation of pads **96** about central axis CA relative to stationary ring **88** is prevented but radial movement of pads **96** relative to stationary ring **88** is permitted. Such (e.g., sliding) engagement of pads **96** with stationary ring **88** may be achieved by way of radial guides between pads and stationary ring **88**. Such radial guides may include respective rails **100** secured to stationary ring **88** and corresponding slots formed in pads **96**. Rails **100** and the corresponding slots may be oriented at least partially radially to permit pads **96** to move radially outwardly to define the expanded configuration E of jaws **32** and to permit pads **96** to move radially inwardly to define the contracted configuration C of jaws **32**. The radial guides may permit radial sliding movement of pads **96** relative to stationary ring **88**. In some embodiments, rails **100** may be integrally formed with stationary ring **88**. Rails **100** may be angularly spaced apart around stationary ring **88**. The number of rails **100** may correspond to the number of pads **96** so that each rail **100** may be associated with a respective pad **96**. In some embodiments, rails **100** may each have a T-shaped cross-sectional profile and the slots may have a corresponding T-shaped cross-sectional profile as well. Rails and slots of other cross-sectional profiles may be suitable. In some embodiments, rails **100** may be secured to or integrally formed with respective pads **96** and the slots may be formed in stationary ring **88** instead.

(47) During operation of jaws **32**, stationary ring **88** may be secured to base **28** and movable ring **94** may be rotatable about central axis CA relative to stationary ring **88**. Movable ring **94** may be coaxial with stationary ring **88** and may also be coaxial with central axis CA. Movable ring **94** may be operatively connected to pads **96** to actuate pads **96** toward expanded configuration E of jaws **32**. Pads **96** may be drivingly connected to movable ring **94** by way of respective cams **104** and follower surfaces **106** so that rotational movement of movable ring **94** may be converted to radially outward movement of pads **96**. Cams **104** may be located radially inwardly of their respective pads

96. Cams **104** may be connected to movable ring **94** via central opening **40** through stationary ring **88** so that cams **104** may be rotatable together with movable ring **94**. The number of cams **104** may correspond to the number of pads **96** so that each cam **104** may be associated with a respective pad **96**. In some embodiments, one or more pads **96** may each be associated with more than one cam **104**.

(48) FIG. **9** is a another perspective view of jaws **32** in isolation. FIG. **9** shows an upper side of jaws **32** on which movable ring **94** is mounted. When stationary ring **88** is secured to base **28**, the actuation of manual lever **34** may be converted to rotational motion of movable ring **94** along arrow **108**. When manual lever **34** is actuated toward the expanded configuration E, the one or more pads **96** of jaws **32** may be urged radially outwardly to define the expanded configuration E of jaws **32**. When manual lever **34** is actuated toward the contracted configuration C, the one or more cams **104** may release the corresponding one or more pads **96** of jaws **32** so that pads **96** may then be manually pushed radially inwardly to define the contracted configuration C of jaws **32**.

(49) In some embodiments, the engagement between pads **96** and stationary ring **88** may define one or more hard stops to limit the radial outward and/or inward movement of pads **96** and thereby prevent pads **96** from becoming disengaged from stationary ring **88** during handling and installation of jaws **32**.

(50) FIGS. **10A** and **10B** are schematic top plan views of one of pads **96** of jaws **32** actuatable using cam **104** and shown in contracted configuration C of jaws **32** and in expanded configuration E of jaws **32**. Cam **104** may be configured to rotate with movable ring **94** and transform rotary motion of cam **104** about central axis CA into linear motion of pad **96** radially outwardly through the engagement of cam **104** with follower surface **106**.

(51) FIGS. **10A** and **10B** show an arcuate path P along which cam **104** travels as cam **104** is rotated about central axis CA along arrow **108**. Since the profile of follower surface **106** does not follow the same path P, this creates an interference between cam **104** and follower surface **106** as cam **104** travels along path P. As shown in FIGS. **10A** and **10B**, the profile of follower surface **106** as viewed along central axis CA may be asymmetric about radial line **110**. Since path P of cam **104** causes cam **104** to intersect pad **96**, the movement of cam **104** and the engagement of cam **104** with follower surface **106** causes pad **96** to be urged radially outwardly toward expanded configuration E. The amount of travel of pad **96** may be selected through the amount of travel of cam **104** and also through the geometric profile of follower surface **106**.

(52) FIGS. **10A** and **10B** show an example of a single-acting cam arrangement where cam **104** pushes pad **96** radially outwardly but does not pull pad **96** radially inwardly when cam **104** is returned to the position in FIG. **10A**. In this scenario, the operator may manually return pads **96** toward the contracted configuration C of jaws **32** when cam **104** is returned to the position in FIG. **10A**. However, some embodiments of jaws **32** may included a double-acting cam arrangement where a cam travels in a slot formed in the pad and the cam is operable to push the pad radially outwardly and also pull the pad radially inwardly by reversing the motion of the cam.

(53) FIG. **11** is a schematic top plan view of an alternate mechanism for actuating one or more pads **96** by the rotation of movable ring **94**. A single linkage **112** is shown in FIG. **11** but embodiments of jaws **32** may include such linkage **112** associated with each pad **96** that must be actuated radially. Linkage **112** may include first link **114A** having a first end pivotally connected to stationary ring **88** or other structure that is stationary relative to movable ring **94**. Linkage **112** may include second link **114B** having a first end pivotally connected to a second end of first link **114A** via pivot pin **116**. Second link **114B** may have a second end that is pivotally connected to pad **96**.

(54) Pivot pin **116** may be connected to movable ring **94** and be rotatable with movable ring **94** along arrow **108** to cause radial movement of pad **96** between contracted configuration C and expanded configuration E of jaws **32**. Linkage **112** may be double acting so that rotation of movable ring **94** in one direction causes radially outward movement of pad **96** toward expanded configuration E, and rotation of movable ring **94** in the opposite direction causes radially inward

movement of pad **96** toward contracted configuration C.

(55) FIG. **12** is a perspective view showing drift **38** being passed through central opening **40** of jaws **32** to access inner support **54** of vane ring **24**. FIG. **12** shows drift **38** being installed onto base **28** of tool **26** after outer ring **48**, vane segments **50**, inner support **54**, heater **30** and jaws **32** have been received on base **28**. Drift **38** may be installed by lowering drift **38** toward base **28** and along central axis CA. Drift **38** may include a lower annular interface for applying axial force **F1** against inner support **54** when components of vane ring **24** are press-fitted together. Once drift **38** has been lowered and installed to engage with inner support **54**, the configuration of tool **26** as shown in FIG. **2** may be achieved. In this configuration, the components of vane ring **24** may be seated at desirable relative positions in preparation for axial force **F1** to be exerted to achieve press fitting of the components of vane ring **24** together.

(56) FIG. **13** is a perspective cross-sectional view of tool **26** of FIG. **2** along line **13-13** of FIG. **2** just prior to the application of axial force **F1** to achieve the press fitting of the components of vane ring **24**. At this stage of assembly, heater **30** may be active to apply heat to outer ring **48** and may have reached a steady state temperature. The application of heat may be sufficient to cause some thermal expansion of outer ring **48** to assist with the subsequent press fitting between outer ring **48**, vane segments **50** and inner support **54**. Heater **30** may be controlled to maintain a desired temperature (or temperature range) of outer ring **48** or of another component that is heated by heater **30**.

(57) At this stage of assembly and while heat is being applied to outer ring **48**, jaws **32** may be in expanded configuration E so that cams **104** are urging pads **96** radially outwardly and consequently, pads **96** are urging vane segments **50** radially outwardly against outer ring **48**. The amount of radial force **F2** that is applied by jaws **32** may only be sufficient to take up slack between the floating vane segment **50** and outer ring **48** and hold the components in place in preparation for the subsequent press fitting.

(58) Vane ring **24** may include seal **118** that is intended to be compressed (i.e., press-fitted) between vane segments **50** and outer ring **48**. Seal **118** may have a generally annular shape extending around central axis CA and may be pre-installed between vane segments **50** and outer ring **48** so that vane segments **50**, outer ring **48** and seal **118** may be received onto base **28** as a unit. Seal **118** may be made from a metallic material and may have a corrugated (e.g., W-shaped) cross-sectional profile. In the expanded configuration E, jaws **32** may take up slack between vane segment **50**, seal **118** and outer ring **48** and hold the components in place in preparation for the subsequent press fitting. In some embodiments, the expansion of jaws **32** may apply a compressive radial pre-loading of seal **118**.

(59) As inner support **54** is lowered and further inserted radially inwardly of vane segments **50**, seal **70** may become disposed and compressed between inner surface **120** of vane segments **50** and an outer surface of inner support **54**. Inner surface **120** may be chamfered or otherwise axially tapered to facilitate the placement of seal **70**. As inner support **54** is further lowered and urged downwardly by drift **38** toward complete installation, a greater radial force **F2** is generated and further compression of seals **70**, **118** is achieved. In some embodiments, only the resilient compression of seals **70**, **118** may solely establish the press fitting between the components of vane ring **24**. In some embodiments, other surfaces within vane ring **24** that are not separated by seals may be in direct press fitting engagement.

(60) Once inner support **54** has been driven to its final position by drift **38**, heater **30** may be deactivated and the assembled vane ring **24** may be allowed to cool. During cooling of vane ring **24**, some of axial force **F1** applied to inner support **54** may be maintained by drift **38** to hold inner support **54** in place for a prescribed time duration. Once sufficient cooling of the assembled (i.e., press-fitted) vane ring **24** has occurred, axial force **F1** may be removed, tool **26** may be dismantled by reversing the steps described herein and vane ring **24** may be withdrawn from base **28**.

(61) Components of tool **26** may be made from metallic materials (e.g., steel) suitable for tooling

applications. In some embodiments, pads **96** may be made from a non-metallic material that can withstand the temperatures of vane ring **24** that are expected during assembly of vane ring **24**. In some embodiments, pads **96** may be made from a polymeric material such as polyether ether ketone (PEEK) for example.

(62) FIG. **14** is a flow diagram of method **200** of assembling vane ring **24** of engine **10** or another vane ring. Method **200** may be performed using tool **26** or other tooling. Method **200** may include components of tool **26** and/or of vane ring **24**. Method **200** may include other actions including other actions disclosed herein. In various embodiments, method **200** may include: receiving outer ring **48** and a plurality of vane segments **50** of vane ring **24** on tooling base **28** where vane segments **50** are disposed radially inwardly of outer ring **48** (block **202**); heating outer ring **48** (block **204**); while outer ring **48** is being heated: urging vane segments **50** radially outwardly toward outer ring **48** (block **206**); partially receiving inner support **54** of vane ring **24** radially inwardly of vane segments **50** (block **208**); and applying axial force **F1** against inner support **54** to install inner support **54** radially inwardly of vane segments **50** (block **210**) and establish a press fit between outer ring **48**, vane segments **50** and inner support **54**.

(63) In some embodiments, the press fitting aspect of the assembly of vane ring **24** may include the compression of one or more seals **70**, **118** between some components of vane ring **24**. For example, in some embodiments, inner support **54** may have seal **70** installed thereon and extending around inner support **54**. Method **200** may include radially compressing seal **70** between inner support **54** and vane segments **50** during the application of axial force **F1** against inner support **54**.

(64) In some embodiments, method **200** may include receiving seal **118** between vane segments **50** and outer ring **48**. Method **200** may include radially compressing seal **118** between vane segments **50** and outer ring **48**. In some embodiments, a radial compressive pre-load may be applied to seal **118** using jaws **32**. In some embodiments, further radial compression of seal **118** may be achieved during the application of axial force **F1** against inner support **54** and the further insertion of inner support **54** centrally of vane segments **50**.

(65) The embodiments described in this document provide non-limiting examples of possible implementations of the present technology. Upon review of the present disclosure, a person of ordinary skill in the art will recognize that changes may be made to the embodiments described herein without departing from the scope of the present technology.

Claims

1. A tool for assembling a vane ring including an outer ring, an inner support and a plurality of vanes disposed radially between the outer ring and the inner support, the tool comprising: a base configured to receive the outer ring and the vanes radially inwardly of the outer ring and angularly distributed about an axis; a heater operable to apply heat to the outer ring when the outer ring is received on the base; jaws having a plurality of radially actuatable pads operable to urge the vanes radially outwardly toward the outer ring; and a press operable to apply an axial force against the inner support to install the inner support radially inwardly of the vanes.
2. The tool as defined in claim 1, wherein the base includes a receptacle configured to receive a boss on a vane segment including one or more of the vanes to angularly locate the vane segment on the base.
3. The tool as defined in claim 2, wherein the receptacle includes a slot extending axially relative to the axis and having an open axial end for receiving the boss therein when the vane segment is received on the base by an axial movement of the vane segment relative to the base.
4. The tool as defined in claim 1, wherein the base includes a locating feature configured to interact with the inner support to angularly locate the inner support on the base.
5. The tool as defined in claim 1, wherein: the heater is configured apply heat to the outer ring by conductive heat transfer; and the heater extends along a majority of a circumference of the outer

ring.

6. The tool as defined in claim 1, wherein: the press includes a drift defining an annular interface for applying the axial force against the inner support; and the jaws define a central opening that is sized to permit a passage of the drift therethrough.

7. The tool as defined in claim 6, wherein the pads are angularly distributed about the central opening of the jaws.

8. The tool as defined in claim 1, wherein the jaws include: a first ring configured to be secured to the base; and a second ring coaxial with the first ring and rotatable about the axis when the first ring is secured to the base, the second ring being operatively connected to the pads to actuate the pads.

9. The tool as defined in claim 8, comprising a plurality of cams connected to the second ring of the jaws and rotatable with the second ring, the cams being respectively engaged with follower surfaces of the pads to convert rotary motion of the cams into radially outward motion of the pads.

10. The tool as defined in claim 8, comprising a cam connected to the second ring of the jaws and rotatable with the second ring, the cam being engaged with a follower surface of a first of the pads to convert rotary motion of the cam into radial motion of the first pad.

11. The tool as defined in claim 10, wherein the cam is located radially inwardly of the first pad.

12. The tool as defined in claim 10, wherein the first pad is engaged with the first ring via a guide extending radially relative to the axis.

13. The tool as defined in claim 12, wherein: the guide includes a rail secured to the first ring; and the guide is slidably received in a slot defined in the first pad.
